

Fundamentals of Artificial Intelligence and Machine Learning

Artificial Intelligence (AI) refers to computer systems engineered to simulate human cognitive functions such as learning, reasoning, and problem-solving.

Machine Learning (ML) is a foundational subset of AI that relies on statistical algorithms to learn patterns directly from data without explicit programming.

Natural Language Processing (NLP) is a specialized branch of AI that enables machines to comprehend, interpret, and generate human language.

Model Evaluation is a critical phase in the ML workflow. During evaluation, models are assessed on unseen test data to measure their performance and generalization capabilities.

Inference refers to the real-world execution phase where a trained AI model makes predictions on new, live input data.

Responsible AI encompasses ethical guidelines, fairness, transparency, accountability, and privacy controls to ensure AI systems are deployed and used responsibly.